Lecture 1

Categories: a theory of systems and processes

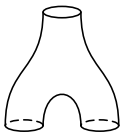
Objectives. *Physics composes processes: one after another, and, from Lecture 3 on, side by side. This lecture fixes the vocabulary that every later lecture uses: categories, functors, natural transformations and equivalences. Its main result says that a representation of a group is a functor, and an intertwiner a natural transformation. Naturality then makes "canonical" precise: $V \to V^{**}$ is natural, and no linear isomorphism $V \to V^{*}$ is.*

By the end of this lecture you will be able to:

- Check the axioms of a category on an example, and read a group as a category with one object.
- Prove that a representation of G is a functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ and an intertwiner a natural transformation, and slide a box through a natural transformation in a picture.
- Prove that $V \to V^{**}$ is natural and that no family of linear isomorphisms $V \to V^{*}$ is, and say why the Riesz map escapes.
- Prove $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$, and explain why the two categories are equivalent but not isomorphic.

1.1 Systems and processes

Three theories from this collection share a shape. In quantum mechanics the systems are Hilbert spaces and the processes linear maps. In representation theory the systems are representations of a group G and the processes are intertwiners (*Symmetries*, Lecture 3). In topology the systems are closed curves, finitely many circles, and a process is a surface whose boundary is the incoming circles and the outgoing ones, a *cobordism*. The pair of pants takes two circles to one:



We draw time upward, as throughout the course: a process starts at the bottom and ends at the top.

In each theory a process has an input and an output. Two processes compose when the output of the first is the input of the second. Composing is associative. Every system has a process that does nothing: the identity map, the identity intertwiner, the cylinder. Baez and Stay set the dictionary out in a table (their Table 2). A theory with this shape is a *category*.

What the vocabulary buys is a way to compare theories. A *functor* translates one category into another, and a *natural transformation* compares two translations. The main result

(Theorem 1.9) says that a representation of G is exactly a functor from a category with one object, and an intertwiner exactly a natural transformation. Two further results show what naturality is for: it makes the word "canonical" a theorem (Theorem 1.11), and it lets two categories be "the same" without being equal (Theorem 1.16).

1.2 Categories and examples

Definition 1.1 (Category)**.** A *category* C consists of

- a collection of *objects* $X, Y, Z, \ldots$;
- for each pair of objects a set $\mathrm{Hom}(X, Y)$ of *morphisms* $f\colon X \to Y$;
- for $f \in \mathrm{Hom}(X, Y)$ and $g \in \mathrm{Hom}(Y, Z)$ a *composite* $g \circ f \in \mathrm{Hom}(X, Z)$;
- for each object an *identity* $\mathrm{id}_X \in \mathrm{Hom}(X, X)$,

such that for all $f\colon X \to Y$, $g\colon Y \to Z$, $h\colon Z \to W$

$$h \circ (g \circ f) = (h \circ g) \circ f,$$
$$f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f.$$

In $g \circ f$ the morphism f acts first. In a picture a morphism $f\colon X \to Y$ is a box on a strand, X below and Y above, and $g \circ f$ puts g above f. A chain of boxes is drawn without brackets:



$$= h \circ (g \circ f) = (h \circ g) \circ f. \tag{1.1}$$

The picture is sound: by associativity, induction on the length shows that all bracketings of a chain give one morphism, and by the unit laws a box id_X is a plain strand (Selinger, Thm. 2.1, drawn there left to right). That is all a picture of a single strand needs. Pictures in two dimensions need more (Lecture 3).

Example 1.2 (Categories)**.** Each of the following is a category.

- Set: sets and functions.
- Vect: complex vector spaces and linear maps; $\mathrm{Vect}_{\mathrm{fd}}$, the finite-dimensional ones.
- $\mathrm{Hilb}_{\mathrm{fd}}$: finite-dimensional complex inner-product spaces and *all* linear maps. Its morphisms are those of $\mathrm{Vect}_{\mathrm{fd}}$.
- BG, for a group G: one object $*$, with

$$\mathrm{Hom}(*, *) = G,$$
$$g \circ h = gh,$$
$$\mathrm{id}_* = 1.$$

  The axioms are the group axioms.

- $\mathrm{Rep}(G)$: the finite-dimensional complex representations (V, ρ_V) of G, and the intertwiners $T : V \to W$,
$$T\rho_V(g) = \rho_W(g)T \quad \text{for all } g \in G$$
(*Symmetries* Def. 3.11, without asking T to be invertible). Composites of intertwiners are intertwiners.
- The *braid groupoid* Br (Mac Lane's braid category): the objects are $n = 0, 1, 2, \ldots$; a morphism $n \to n$ is a braid of n strands up to isotopy with its ends fixed, and there is none from m to $n \neq m$. Braids compose by stacking, the second on top:



- Cob_2: the objects are finite disjoint unions of oriented circles, a morphism is an oriented surface between them up to diffeomorphism fixing the boundary, composition glues outgoing circles to incoming ones, and the identity is the cylinder (Kock, 1.3.14–1.3.20, for the details). Until Lecture 14 it is used only as pictures.

Definition 1.3 (Isomorphism, groupoid)**.** A morphism $f : X \to Y$ is an *isomorphism* if some $g : Y \to X$ has
$$g \circ f = \mathrm{id}_X,$$
$$f \circ g = \mathrm{id}_Y .$$
Then g is unique, written f^{-1}, and $X \cong Y$. A *groupoid* is a category in which every morphism is an isomorphism.

Uniqueness: if h is another such morphism, then
$$h = h \circ (f \circ g) = (h \circ f) \circ g = g.$$

The isomorphisms of Vect are the invertible linear maps. BG is a groupoid, and so is Br: the inverse of a braid is its mirror image in a horizontal line. The finite-dimensional Hilbert spaces with only the unitary maps form a groupoid; with all linear maps they form $\mathrm{Hilb}_{\mathrm{fd}}$, which is not one.

1.3 Functors

Definition 1.4 (Functor)**.** A *functor* $F : C \to \mathcal{D}$ assigns to each object X of C an object $F(X)$ of $\mathcal{D}$, and to each $f : X \to Y$ a morphism $F(f) : F(X) \to F(Y)$, such that
$$F(g \circ f) = F(g) \circ F(f),$$
$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)} .$$
The *opposite category* C^{op} has the objects of C and a morphism $X \to Y$ for each morphism $Y \to X$ of C, composed in the reverse order. A functor $C^{\mathrm{op}} \to \mathcal{D}$ is a *contravariant* functor from C to $\mathcal{D}$: it reverses composition.

Example 1.5 (Functors)**.** Each of the following is a functor.

- The *forgetful* functor $\mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$ sends (V, ρ_V) to V and an intertwiner to itself.

- The *dual*: $V^* = \mathrm{Hom}(V, \mathbb{C})$, and $f : V \to W$ goes to $f^* : W^* \to V^*$,
$$f^*(\varphi) = \varphi \circ f$$
(*Linear Algebra*'s dual map T', Def. 11.20). For $g : W \to Z$,
$$\begin{aligned}(g \circ f)^*(\varphi) &= \varphi \circ g \circ f \\ &= f^*\big(g^*(\varphi)\big),\end{aligned}$$
so
$$(g \circ f)^* = f^* \circ g^*,$$
and the dual is a functor $\mathrm{Vect}^{\mathrm{op}} \to \mathrm{Vect}$. Applied twice it is covariant: $V \mapsto V^{**}$, $f \mapsto f^{**}$.
- *Forgetting crossings*: a braid on n strands sends the strand that starts at i to a position $\pi(i)$, and $\pi \in S_n$. This defines a functor from Br to the groupoid with objects n and morphisms $\mathrm{Hom}(n, n) = S_n$. The braid drawn above gives $1 \mapsto 3$, $2 \mapsto 1$, $3 \mapsto 2$. *Symmetries* (Lecture 1) draws permutations as such strands, read upward as here.

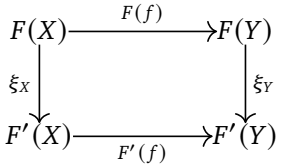
Lemma 1.6. *A functor sends isomorphisms to isomorphisms, with $F(f^{-1}) = F(f)^{-1}$.*

Proof.
$$\begin{aligned}F(f^{-1}) \circ F(f) &= F(f^{-1} \circ f) \\ &= F(\mathrm{id}_X) \\ &= \mathrm{id}_{F(X)},\end{aligned}$$
and likewise in the other order. ∎

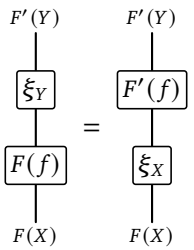
1.4 Natural transformations; $\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}})$

Definition 1.7 (Natural transformation)**.** Let $F, F' : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\xi : F \Rightarrow F'$ is a morphism $\xi_X : F(X) \to F'(X)$ for each object X, its *component*, such that for every $f : X \to Y$
$$F'(f) \circ \xi_X = \xi_Y \circ F(f).$$
It is a *natural isomorphism* if every component is an isomorphism.

The condition says that the square



commutes: the two paths from $F(X)$ to $F'(Y)$ agree. Drawn upward, it says that the box $F(f)$ slides through ξ and comes out as $F'(f)$:



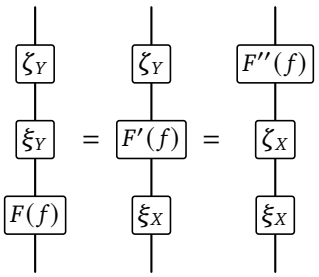
(1.2)

The square names the objects at its corners; the picture puts them on strands and lets the boxes move. From Lecture 3 on we use the picture, because it extends to two dimensions.

Lemma 1.8 (Composing natural transformations). *If $\xi\colon F \Rightarrow F'$ and $\zeta\colon F' \Rightarrow F''$ are natural, so is $\zeta \circ \xi\colon F \Rightarrow F''$, with components*

$$(\zeta \circ \xi)_X = \zeta_X \circ \xi_X.$$

Proof. Slide $F(f)$ through ξ by (1.2), then $F'(f)$ through ζ by (1.2) for ζ:



(1.3)

The outer pictures are the two sides of naturality for $\zeta \circ \xi$. ∎

If the objects of C form a set, as for BG, the functors $C \to \mathcal{D}$ form a category $\mathrm{Fun}(C, \mathcal{D})$: its morphisms are the natural transformations, composed by Lemma 1.8, with identities $(\mathrm{id}_F)_X = \mathrm{id}_{F(X)}$. Associativity and the unit laws hold component by component.

Two categories are *isomorphic*, $C \cong \mathcal{D}$, if there are functors $C \to \mathcal{D} \to C$ whose two composites are the identity functors.

Theorem 1.9 (Representations are functors). *A functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ is a representation of G, and a natural transformation between two such functors is an intertwiner. Hence*

$$\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}}),$$

an isomorphism of categories that changes no data.

Proof. A functor $F\colon BG \to \mathrm{Vect}_{\mathrm{fd}}$ is a space $V = F(*)$ and a linear map $F(g)\colon V \to V$ for each $g \in G$. Since $g \circ h = gh$ in BG, the axioms read

$$F(gh) = F(g)F(h),$$
$$F(1) = \mathrm{id}_V\,.$$

By Lemma 1.6 each $F(g)$ is invertible, so $\rho = F$ is a homomorphism $G \to \mathrm{GL}(V)$; conversely, every representation satisfies the two axioms.

A natural transformation $\xi\colon F \Rightarrow F'$ has the single component $T = \xi_*\colon V \to V'$, and naturality for the morphism g reads

$$\rho'(g)\,T = T\rho(g).$$

The map T is an intertwiner, and every intertwiner is natural. Natural transformations compose by composing components, as intertwiners do, and the identities agree. ∎

Example 1.10 (S_3 on $\mathbb{C}^3$). Let S_3 permute the coordinates, $\rho(\pi)e_i = e_{\pi(i)}$, and let $\mathbb{C}$ carry the trivial representation, $\pi \mapsto 1$. The sum $s(x) = x_1 + x_2 + x_3$ is natural from ρ to the trivial

representation, since permuting the coordinates does not change their sum. The coordinate $p_1(x) = x_1$ is linear but not natural: for the transposition $(1\,2)$,

$$p_1\big(\rho((1\,2))e_1\big) = p_1(e_2) = 0,$$
$$p_1(e_1) = 1 \neq 0.$$

So the forgetful functor $\mathrm{Rep}(S_3) \to \mathrm{Vect}_{\mathrm{fd}}$ misses morphisms. In the other direction, $u(1) = e_1 + e_2 + e_3$ is natural from the trivial representation to ρ. By Lemma 1.8 so is the composite $u \circ s$, whose matrix has every entry 1: indeed it commutes with every permutation matrix.

1.5 Equivalence; V^{**} vs V^{*}

Linear Algebra (Thm. 11.17) calls $V \to V^{**}$ canonical because it uses no basis; naturality makes this a theorem. To compare V with V^{*}, we must first make the dual covariant. On the groupoid of finite-dimensional spaces and linear isomorphisms, set

$$D(V) = V^{*},$$
$$D(f) = (f^{*})^{-1}.$$

This is a functor, since

$$\big((g \circ f)^{*}\big)^{-1} = (g^{*})^{-1} \circ (f^{*})^{-1}.$$

Theorem 1.11 (The double dual is natural; the dual is not)**.**

(a) The maps $a_V : V \to V^{**}$,

$$a_V(v)(\varphi) = \varphi(v),$$

form a natural transformation $\mathrm{id}_{\mathrm{Vect}} \Rightarrow (-)^{**}$, and a natural isomorphism on $\mathrm{Vect}_{\mathrm{fd}}$.

(b) No family of linear isomorphisms $\theta_V : V \to V^{*}$, one for each finite-dimensional V, is a natural transformation from the identity to D. On $\mathrm{Hilb}_{\mathrm{fd}}$ none is natural even for the unitary maps alone.

Proof. (a) For $f : V \to W$ the double dual map is

$$f^{**}(x) = x \circ f^{*}.$$

For $v \in V$ and $\psi \in W^{*}$,

$$\begin{aligned} f^{**}\big(a_V(v)\big)(\psi) &= a_V(v)(\psi \circ f) \\ &= \psi(f(v)) \\ &= a_W(f(v))(\psi), \end{aligned}$$

so

$$f^{**} \circ a_V = a_W \circ f.$$

For finite-dimensional V, a_V is an isomorphism (*Linear Algebra* Thm. 11.17).

(b) Naturality for $f : V \to W$ reads

$$\theta_W \circ f = (f^{*})^{-1} \circ \theta_V,$$

that is,

$$f^{*} \circ \theta_W \circ f = \theta_V.$$

Take $V \neq 0$ and $f = i\,\mathrm{id}_V$, which is unitary for every inner product. Then $f^{*} = i\,\mathrm{id}_{V^{*}}$, and since θ_V is linear,

$$f^{*} \circ \theta_V \circ f = i \cdot i\,\theta_V = -\theta_V.$$

So naturality demands $-\theta_V = \theta_V$, that is $\theta_V = 0$, which is not an isomorphism. ∎

Example 1.12 (A basis is a choice)**.** A basis (e_i) of V does give an isomorphism $\theta\colon V \to V^*$, $\theta(e_i) = e^i$, onto the dual basis. Take instead the basis $e'_i = 2e_i$. Its dual basis is $e'^{\,i} = e^i/2$, and its isomorphism θ' sends $e_i = e'_i/2$ to $e'^{\,i}/2 = e^i/4$:

$$\theta' = \tfrac{1}{4}\,\theta.$$

The isomorphism changes with the basis, and (b) says more: no way of choosing a basis for every space, however clever, gives a natural family.

Example 1.13 (The Riesz map is not a counterexample)**.** On $\mathrm{Hilb}_{\mathrm{fd}}$ the Riesz map

$$R_V(u) = \langle u, \cdot \rangle$$

is a bijection $V \to V^*$, and for every unitary f

$$f^*\big(R_W(fu)\big) = \langle fu, f\,\cdot\rangle = R_V(u).$$

It escapes (b) because the inner product is antilinear in its first slot (*Linear Algebra* Prop. 11.13), so that

$$R_V(iu) = -i\,R_V(u).$$

In the test of (b) the two factors i cancel instead of multiplying to -1:

$$\langle iu, iw\rangle = (-i)\,i\,\langle u, w\rangle = \langle u, w\rangle\,.$$

Hence R_V is not a morphism of Vect. Lecture 4 builds duals from ev and coev, never from an inner product.

Theorem 1.11 compares objects inside one category. The last question compares two categories. $\mathrm{Vect}_{\mathrm{fd}}$ has a whole class of objects, all the spaces; matrices need only the numbers $0, 1, 2, \ldots$ Linear algebra treats the two as interchangeable, and the next definition says in what sense.

Definition 1.14 (Equivalence)**.** An *equivalence* $C \simeq \mathcal{D}$ is a pair of functors $F\colon C \to \mathcal{D}$ and $G\colon \mathcal{D} \to C$ with natural isomorphisms

$$G \circ F \cong \mathrm{id}_C,$$
$$F \circ G \cong \mathrm{id}_{\mathcal{D}}\,.$$

Definition 1.15 (Full, faithful, essentially surjective)**.** A functor $F\colon C \to \mathcal{D}$ is *faithful* or *full* if each map

$$\mathrm{Hom}(X, Y) \to \mathrm{Hom}(F(X), F(Y))$$

is injective or surjective, respectively. It is *essentially surjective* if every object of $\mathcal{D}$ is isomorphic to some $F(X)$.

The forgetful functor of Example 1.10 is faithful but not full: it keeps every intertwiner, and misses p_1. Let $\mathrm{Mat}_{\mathbb{C}}$ be the category with objects $n = 0, 1, 2, \ldots$, whose morphisms $m \to n$ are the complex $n \times m$ matrices,

$$\mathrm{Hom}(m, n) = \mathbb{C}^{n \times m},$$

composed by the matrix product. It is *skeletal*: isomorphic objects are equal, because an invertible matrix is square. A skeletal category equivalent to C is a *skeleton* of C.

Theorem 1.16 (Vector spaces are matrices, up to equivalence). *The functor* $M\colon \mathrm{Mat}_{\mathbb{C}} \to \mathrm{Vect}_{\mathrm{fd}}$,

$$M(n) = \mathbb{C}^n,$$
$$M(A) = (x \mapsto Ax),$$

is an equivalence. So $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$*, and* $\mathrm{Mat}_{\mathbb{C}}$ *is a skeleton of* $\mathrm{Vect}_{\mathrm{fd}}$.

Proof. Choose for every V a basis, as an isomorphism $b_V\colon \mathbb{C}^{\dim V} \to V$, with $b_{\mathbb{C}^n} = \mathrm{id}$. Set

$$G(V) = \dim V,$$
$$G(f) = b_W^{-1} \circ f \circ b_V,$$

the matrix of $f\colon V \to W$. For $g\colon W \to X$ the b's cancel,

$$\begin{aligned} G(g) \circ G(f) &= b_X^{-1} \circ g \circ b_W \\ &\quad \circ b_W^{-1} \circ f \circ b_V \\ &= G(g \circ f), \end{aligned}$$

so G is a functor, and $G \circ M = \mathrm{id}$ because $b_{\mathbb{C}^n} = \mathrm{id}$. The components b_V form a natural isomorphism $M \circ G \Rightarrow \mathrm{id}$: by the definition of $G(f)$,

$$f \circ b_V = b_W \circ MG(f).$$

■

Proposition 1.17 (· *sketched*). *A functor that is full, faithful and essentially surjective is an equivalence.*

The proof is the one above, with choices in place of bases. For each object Y of $\mathcal{D}$ choose an object $G(Y)$ and an isomorphism $b_Y\colon F(G(Y)) \to Y$. For $g\colon Y \to Y'$, fullness and faithfulness give exactly one morphism $G(g)$ with

$$F(G(g)) = b_{Y'}^{-1} \circ g \circ b_Y,$$

and b is then a natural isomorphism $F \circ G \Rightarrow \mathrm{id}$. For the other one, full faithfulness lifts each $b_{F(X)}^{-1}$ to an isomorphism $X \to G(F(X))$, natural in X (Leinster, Prop. 1.3.18 and Ex. 1.3.32). Like the bases, the choices are the price of an equivalence; an isomorphism of categories needs none.

Example 1.18 (Equivalent, not isomorphic). $\mathrm{Vect}_{\mathrm{fd}}$ and $\mathrm{Mat}_{\mathbb{C}}$ are not isomorphic. An isomorphism of categories is injective on objects and keeps isomorphic objects isomorphic (Lemma 1.6). The line $\{(z, 0)\} \subset \mathbb{C}^2$ and $\mathbb{C}$ are different, isomorphic objects of $\mathrm{Vect}_{\mathrm{fd}}$; in the skeletal $\mathrm{Mat}_{\mathbb{C}}$ no two different objects are isomorphic. So categories are compared up to equivalence: Lecture 2 finds $\mathrm{Rep}(S_3)$ equivalent to three copies of $\mathrm{Vect}_{\mathrm{fd}}$, and Lecture 3 replaces every monoidal category by an equivalent strict one.

Summary.

- *Handed on:* categories, functors, natural transformations and equivalences; pictures with time running upward, in which a chain of boxes needs no brackets, (1.1).
- *The main result:* a representation is a functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ and an intertwiner a natural transformation, so $\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}})$ (Theorem 1.9); in pictures, naturality is sliding, (1.2).

- *Canonical, made precise:* $V \to V^{**}$ is natural, and no linear isomorphism $V \to V^{*}$ is, even for unitaries (Theorem 1.11). The Riesz map escapes only by being antilinear (Example 1.13).
- *The guard:* $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$, but the two are not isomorphic (Theorem 1.16 and Example 1.18).
- *Next:* Lecture 2 characterises $\oplus$ and $\otimes$ by universal properties, and shows why $\mathrm{Rep}(G)$ looks like copies of $\mathrm{Vect}_{\mathrm{fd}}$.

References

Leinster, *Basic Category Theory*: categories, Def. 1.1.1, p. 10, and brackets, Rem. 1.1.2, pp. 10–11; functors, Def. 1.2.1, p. 17, and contravariance, Def. 1.2.10, p. 22; representations as functors, Ex. 1.2.8, p. 22; natural transformations, Def. 1.3.1, p. 28; the double dual, Ex. 1.3.14, pp. 32–33; equivalence, Def. 1.3.15 and Prop. 1.3.18, p. 34; the functor back, Ex. 1.3.32, and Mat, Ex. 1.3.33, p. 40. Mac Lane, *Categories for the Working Mathematician*: groupoids, §I.5, p. 20; skeletons, §IV.4, p. 93; the braid category, §XI.4, pp. 260–262. Heunen and Vicary, *Categories for Quantum Theory*, Ch. 0: intertwiners as natural transformations, Ex. 0.20, p. 10; pictures drawn upward, §0.1.5, pp. 8–9; $\mathrm{Mat}_{\mathbb{C}} \simeq \mathrm{Vect}_{\mathrm{fd}}$, Prop. 0.37, p. 15. Selinger, *A survey of graphical languages for monoidal categories*, arXiv:0908.3347, §2, pp. 5–6. Baez and Stay, *Physics, topology, logic and computation: a Rosetta Stone*, arXiv:0903.0340, §2, Table 2, p. 4. Kock, *Frobenius Algebras and 2D Topological Quantum Field Theories*, §1.3, pp. 42–45.